Zoom X Outlook Integration
Setup and Test Documentation

Project	Zoom X Outlook
Current Status	Working in browser and inside Zoom Workplace; Outlook connection and last 3 email fetch verified.

1. Objective

The goal of this work was to build a custom Zoom application that connects to Microsoft Outlook without using Zoom's built-in Outlook integration. The app had to authenticate a user with Microsoft, fetch the latest three emails through Microsoft Graph, and display them inside the Zoom app UI.

2. Final Architecture

- Zoom App: hosts the UI inside Zoom Workplace.
- Node.js + Express backend: serves the frontend, handles Microsoft OAuth, proxies the Zoom Apps SDK, and fetches emails from Microsoft Graph.
- Microsoft Entra / Azure App Registration: provides delegated Outlook permissions and OAuth credentials.
- ngrok: exposes the local backend over HTTPS for Zoom and Microsoft redirect callbacks.
- Link-ID based connection mapping: used to bridge the external browser OAuth flow back to the Zoom embedded webview.

3. What Was Created

- A Zoom app named "Zoom X Outlook".
- A Microsoft Entra app registration for Outlook access.
- A local Node.js backend project.
- A static frontend served from the backend.
- A working in-Zoom email display flow.

4. Key Configuration

4.1 Zoom App

- App type: User-managed Zoom App.
- OAuth Redirect URL: https://provider-vivacious-greyhound.ngrok-free.dev/auth/zoom/callback
- OAuth Allow List: https://provider-vivacious-greyhound.ngrok-free.dev/auth/zoom/callback
- Home URL: https://provider-vivacious-greyhound.ngrok-free.dev
- Domain Allow List: provider-vivacious-greyhound.ngrok-free.dev
- Surface enabled: Meetings
- In-Client OAuth enabled

- Zoom App SDK enabled with openUrl capability

4.2 Microsoft Entra / Azure

- Redirect URI: https://provider-vivacious-greyhound.ngrok-free.dev/auth/outlook/callback
- Supported accounts: Any Entra ID tenant + personal Microsoft accounts
- Delegated permissions used for the working build: Mail.Read, offline_access, openid, profile, email
- Authentication flow: Authorization Code flow through @azure/msal-node

5. Local Backend Setup

- Project folder: zoom-outlook-backend
- Packages installed: express, express-session, dotenv, @azure/msal-node, node-fetch initially, later native fetch used
- Runtime tested successfully on Node.js v25.9.0
- Frontend file served from public/index.html
- Backend entry file: server.js

6. Core Implementation Flow

1. Run the Express backend locally on port 3000.

2. Expose the app through ngrok so Zoom and Microsoft can reach it over HTTPS.

3. Load the app in Zoom via the Home URL.

4. Use Zoom App SDK openUrl to launch Microsoft sign-in in an external browser.

5. Complete Microsoft OAuth in the browser.

6. Store the Outlook connection on the backend using a generated linkId instead of relying only on cookies.

7. Poll the backend from the Zoom app until the Outlook connection is marked as complete.

8. Fetch the latest three emails with Microsoft Graph and show them in the Zoom UI.

7. Important Technical Decisions

7.1 Why Zoom OAuth callback was added

A simple /auth/zoom/callback route was added so the Zoom installation and authorization redirect could complete cleanly during local testing. For the working version, the project did not require exchanging the Zoom code for a Zoom access token because the actual business requirement was Outlook access, not Zoom REST API access.

7.2 Why the Zoom App SDK was needed

Inside Zoom Workplace, clicking a normal browser redirect or window.open was not enough to launch the external Microsoft OAuth flow reliably. The Zoom App SDK openUrl capability was enabled so the app could open the Outlook sign-in URL correctly from the Zoom embedded client.

7.3 Why session-only auth did not work inside Zoom

When Microsoft sign-in completed in Chrome, the resulting cookie and session belonged to the external browser context. The Zoom embedded webview did not share that same session state, so the Zoom app still appeared disconnected. This was solved by introducing a linkId-based connection record stored on the backend.

8. Security and Response Header Work

- Added required headers for Zoom Home URL validation, including Strict-Transport-Security, X-Content-Type-Options, Content-Security-Policy, and Referrer-Policy.
- Expanded CSP rules so the Zoom Apps SDK and required Microsoft and Zoom endpoints could load correctly.
- Added a same-origin /zoom-sdk.js route that proxies the Zoom SDK script, which resolved SDK loading issues inside the Zoom webview.

9. Main Backend Endpoints

Endpoint	Purpose
/	Serves the frontend UI
/zoom-sdk.js	Proxies Zoom Apps SDK through the same origin
/auth/outlook/start?linkId=...	Starts Microsoft OAuth flow
/auth/outlook/callback	Receives Microsoft OAuth callback and stores connection
/auth/outlook/status?linkId=...	Checks whether Outlook is connected for the linkId
/emails/latest?linkId=...	Fetches latest three emails using Microsoft Graph
/auth/zoom/callback	Handles Zoom install callback for local testing
/debug/sdk	Simple backend health and debug endpoint

10. Frontend Behavior

- Generates and stores a persistent linkId in localStorage.
- Initializes the Zoom SDK with the openUrl capability.
- Uses zoomSdk.openUrl to launch Microsoft sign-in.
- Polls /auth/outlook/status until the connection is complete.
- Calls /emails/latest with linkId and renders the returned messages in the Zoom UI.
- Includes a small debug view for troubleshooting SDK and connection state.

11. Testing Results

- Browser test passed: Outlook connection worked and latest three emails were fetched.
- Zoom UI load test passed: app loaded inside Zoom Workplace under the Apps section.
- Zoom SDK validation passed: window.zoomSdk became available and config succeeded.
- External OAuth launch passed: Connect Outlook opened Microsoft sign-in from the Zoom app.
- Final end-to-end Zoom test passed: latest three emails displayed successfully inside Zoom Workplace.

12. Known Limitations of the Local Test Build

- The Outlook connection store is currently in-memory, so restarting the Node server clears connections.
- The implementation is suitable for local development and proof of concept, not for production token storage.
- Client secrets should be rotated if they were ever exposed during setup or testing.
- A production version should use a database or secure token store, structured logging, and stronger secret handling.

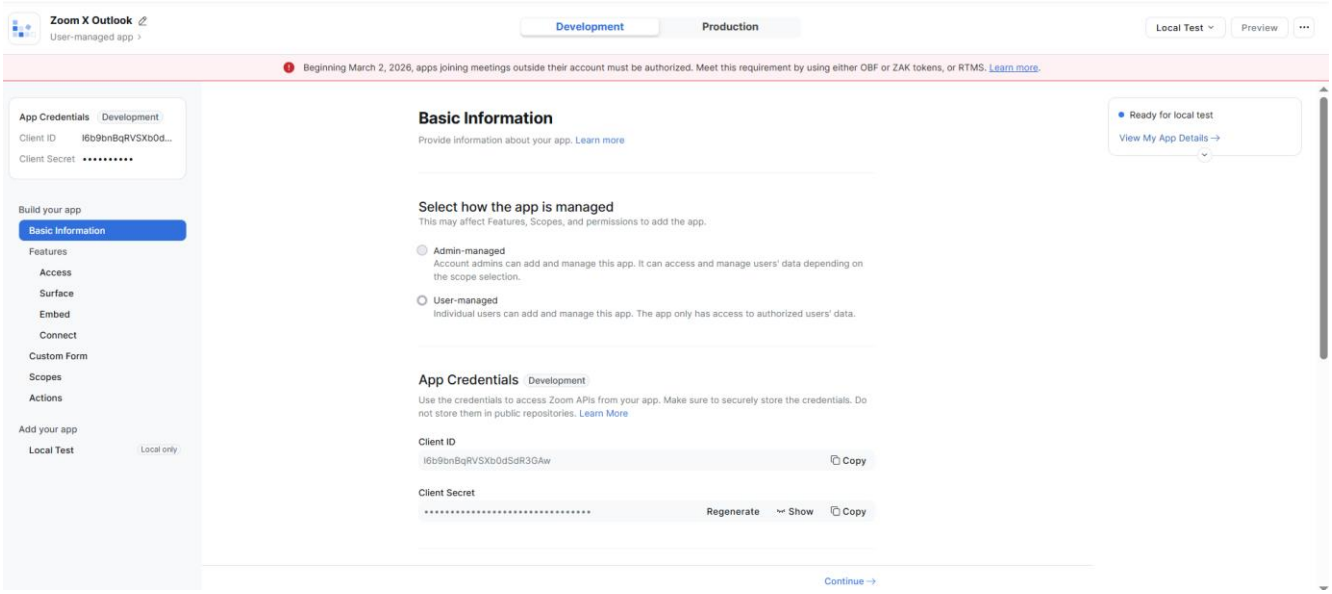
14. Local Run Checklist

- Start the backend with node server.js.
- Start ngrok with ngrok http 3000.
- Open Zoom Workplace and launch the Zoom X Outlook app.
- Click Connect Outlook and complete Microsoft sign-in in the external browser.
- Return to Zoom and click Fetch last 3 emails.
- Verify that the latest three emails are displayed in the Zoom app UI.

15. Final Outcome

The custom Zoom X Outlook integration is working correctly in the current local development environment. The app successfully opens Outlook sign-in, stores the connection using a linkId-based backend mapping, and retrieves the latest three emails for display inside Zoom Workplace.

ZOOM Marketplace Screenshots



App Credentials Development

Client ID I6b9bnBqRVSXb0d...

Client Secret ••••••••••

Build your app

- Basic Information
- Features
  - Access
  - Surface
  - Embed
  - Connect
- Custom Form
- Scopes
- Actions

Add your app

- Local Test Local only

Surface

These features dictate what can be seen in the Zoom client. Choose the features you want to enable for your application, but please note that not all features may be supported on all Zoom products. Learn more

Home URL

The Home URL enables you to run a web browser inside the Zoom client and display the Home URL domain inside the in-client web browser.

https://provider-vivacious-greyhound.ngrok-free.dev

Domain Allow List

URLs added here are to be accepted by the web browser running in the native Zoom Client.

provider-vivacious-greyhound.ngrok-free.dev

This domain requires review. Briefly explain your reason for using it.

for testing purpose

+ Add Domain

Select where to use your app New

Select the Zoom products that this app works in. Users can open and use the app directly from these products.

Note: Some features may not be available depending on the products you select.

Meetings | Webinars | Rooms | Phone

Zoom App SDK

Select the APIs and events that you want to use in the Zoom client app.Learn more

+ Add APIs 1 Selected

Works in:

Guest Mode

A special mode that allows your app to create a delightful, successful experience for unauthenticated or restricted users who are joining meetings or webinars. Learn More

Works in:

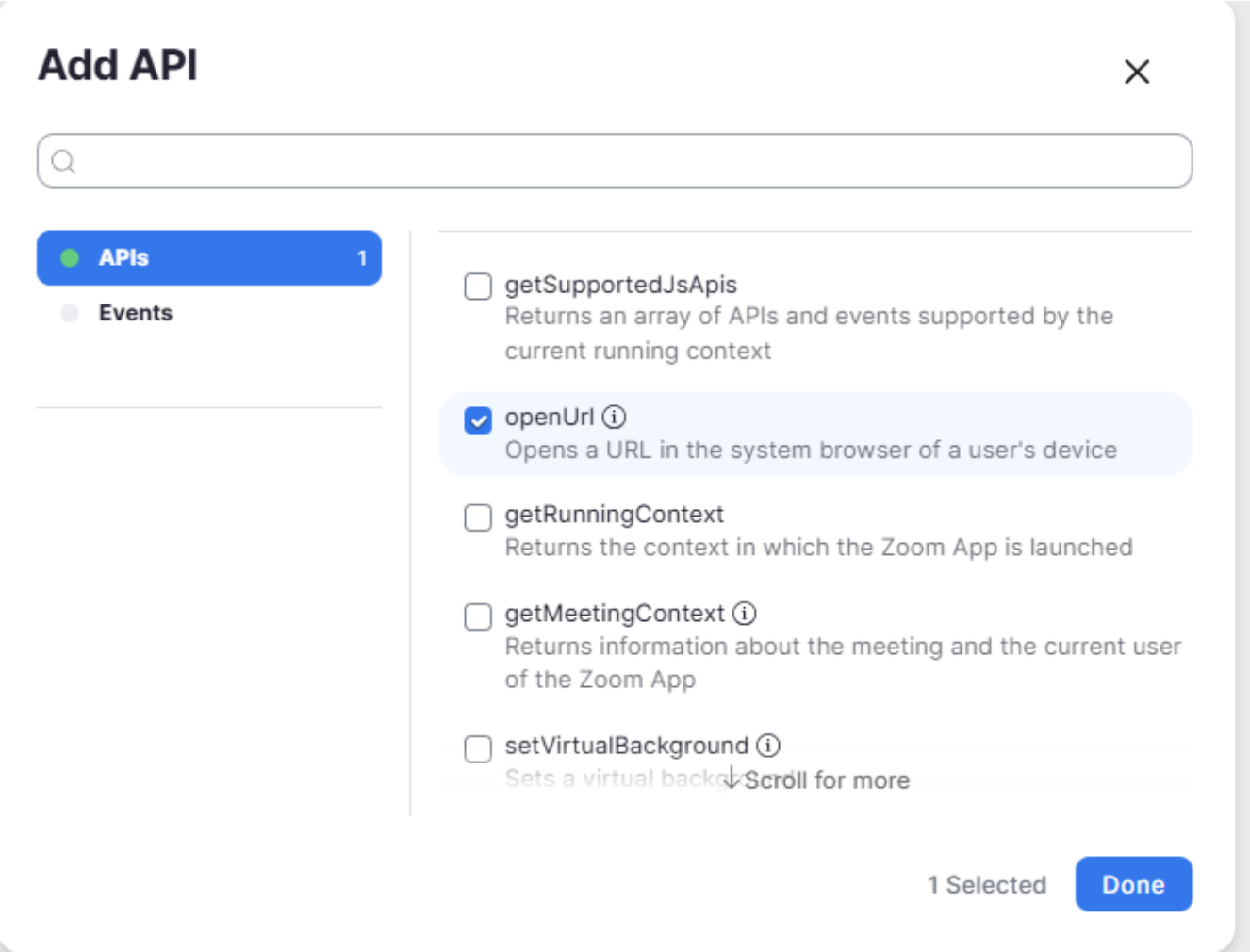
In-Client OAuth

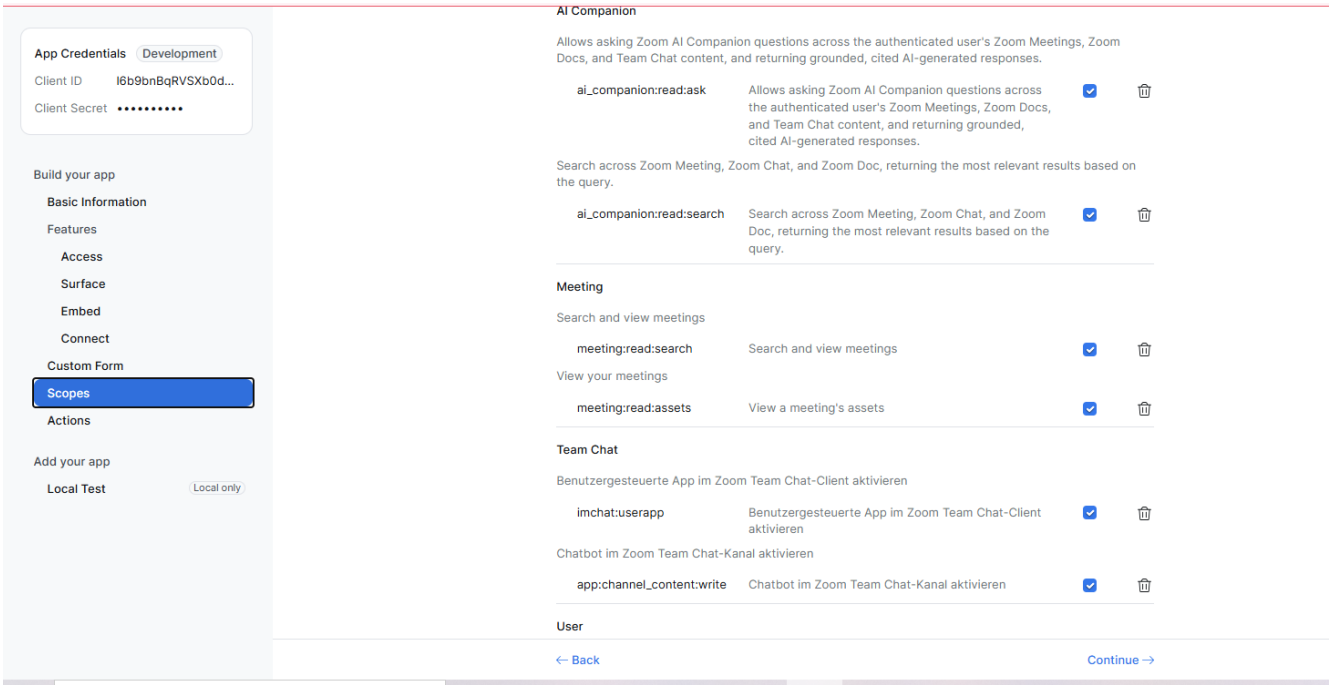
Users can complete the authorization in Zoom Client without opening system browser. It requires the application to support the in-client authorization flow. Learn more

Works in:

Collaborate Mode

← Back Continue →


Add API
APIs 1
Events
getSupportedJsApis
Returns an array of APIs and events supported by the current running context
openUrl
Opens a URL in the system browser of a user's device
getRunningContext
Returns the context in which the Zoom App is launched
getMeetingContext
Returns information about the meeting and the current user of the Zoom App
setVirtualBackground
Scroll for more
1 Selected
Done


App Credentials Development
Client ID l6b9bnBqRVSXb0d...
Client Secret ••••••••••
Build your app
Basic Information
Features
Access
Surface
Embed
Connect
Custom Form
Scopes
Actions
Add your app
Local Test Local only
AI Companion
Allows asking Zoom AI Companion questions across the authenticated user's Zoom Meetings, Zoom Docs, and Team Chat content, and returning grounded, cited AI-generated responses.
ai_companion:read:ask
Allows asking Zoom AI Companion questions across the authenticated user's Zoom Meetings, Zoom Docs, and Team Chat content, and returning grounded, cited AI-generated responses.
Search across Zoom Meeting, Zoom Chat, and Zoom Doc, returning the most relevant results based on the query.
ai_companion:read:search
Search across Zoom Meeting, Zoom Chat, and Zoom Doc, returning the most relevant results based on the query.
Meeting
Search and view meetings
meeting:read:search
Search and view meetings
View your meetings
meeting:read:assets
View a meeting's assets
Team Chat
Benutzergesteuerte App im Zoom Team Chat-Client aktivieren
imchat:userapp
Benutzergesteuerte App im Zoom Team Chat-Client aktivieren
Chatbot im Zoom Team Chat-Kanal aktivieren
app:channel_content:write
Chatbot im Zoom Team Chat-Kanal aktivieren
User
← Back
Continue →

Zoom X Outlook
User-managed app ›

Development | Production

Beginning March 2, 2026, apps joining meetings outside their account must be authorized. Meet this requirement by using either OBF or ZAK tokens, or

App Credentials Development
Client ID l6b9bnBqRVSXb0d...
Client Secret •••••••••

Build your app
- Basic Information
- Features
  - Access
  - Surface
  - Embed
  - Connect
- Custom Form
- Scopes
- Actions

Add your app
- Local Test (Local only)

Zoom Docs

Add collaborators for a file		
docs:write:collaborator	Add collaborators for a file	☑
Create a file export		
docs:write:export	Create a file export	☑
Create a file upload		
docs:write:file_uploads	Create a file upload	☑
Create new file by import		
docs:write:import	Create a new file by import	☑
Data table row read scope		
docs:read:data_table_row	Data table row read scope	☑
Delete a file		
docs:delete:file	Delete a file	☑
Edit basic info of a file		
docs:update:file	Edit basic info of a file	☑
docs:write:file	Create a file	☑
Get basic info of the file		
docs:read:list_children	List basic info of the children of a file	☑
docs:read:file	Get basic info of a specific file	☑
Get collaborators of a file		
docs:read:list_file_collabor	Get collaborators of a file	☑

← Back Continue →

Zoom X Outlook
User-managed app ›

Development | Production

Beginning March 2, 2026, apps joining meetings outside their account must be authorized. Meet this requirement by using either OBF or ZAK tokens, or RTMS. Learn m

App Credentials Development
Client ID l6b9bnBqRVSXb0d...
Client Secret •••••••••

Build your app
- Basic Information
- Features
  - Access
  - Surface
  - Embed
  - Connect
- Custom Form
- Scopes
- Actions

Add your app
- Local Test (Local only)

docs:read:export	Get file export status	☑
Get general access setting of a file		
docs:read:general_access	Get general access setting of a file	☑
Get import task status		
docs:read:import	Get import task status	☑
List data tables in a file		
docs:read:data_table	List data tables in a file	☑
Modify a collaborator's role in a file		
docs:update:collaborator	Modify a collaborator's role in a file	☑
Remove collaborators from a file		
docs:delete:collaborator	Remove collaborators from a file	☑
Transfer ownership of a file		
docs:update:file_owner	Transfer ownership of a file	☑
Update general access of a file		
docs:update:general_access	Update general access of a file	☑

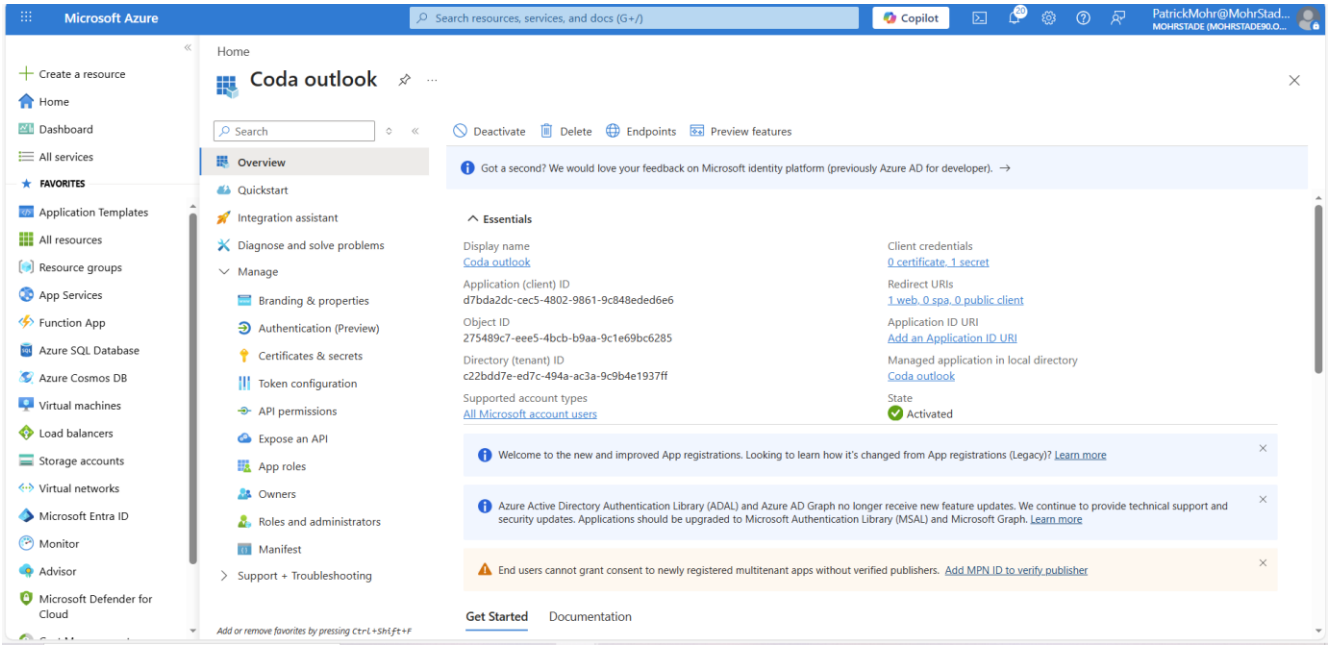
Zoom Events

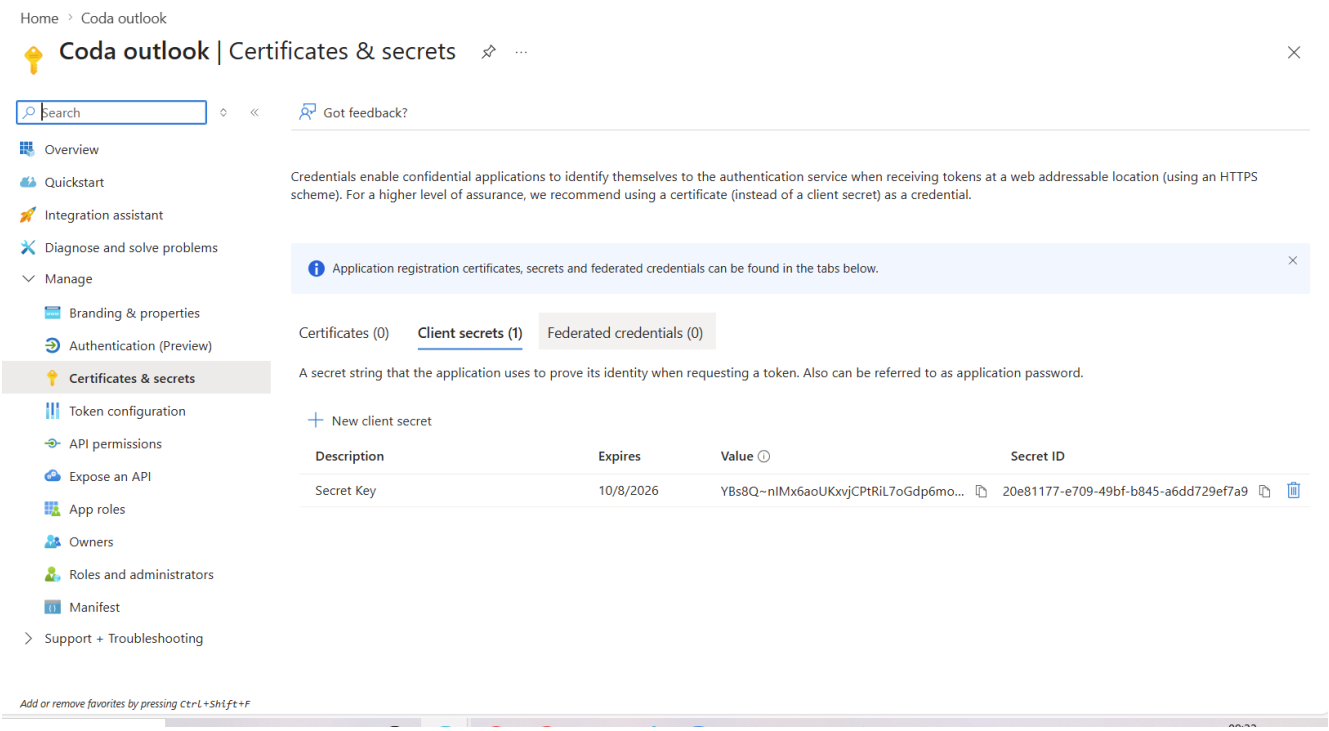
Event anzeigen		
zoom_events:read:event	Event anzeigen	☑

Scope description
Specify how this data is used within the application, and whether or not the data is stored. If stored,

← Back Continue →

Microsfot Azure





Home › Coda outlook

Coda outlook | Authentication (Preview)

Search

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- Diagnose and solve problems
- Manage
  - Branding & properties
  - Authentication (Preview)
  - Certificates & secrets
  - Token configuration
  - API permissions
  - Expose an API
  - App roles
  - Owners
  - Roles and administrators
  - Manifest
- Support + Troubleshooting

Add or remove favorites by pressing Ctrl+Shift+F

Got feedback?

Welcome to the new and improved experience for authentication. To switch to the old experience, please click here.

Redirect URI configuration | Supported accounts | Settings

A redirect URI, or reply URL, is the location where the Entra authorization server sends the user and delivers tokens once the user has successfully signed in or signed out. Learn more about what is a redirect URI

Add Redirect URI | Delete

If you want to view or change implicit grant and hybrid flows settings or front-channel logout URL, please visit the settings tab.

Start typing a reply url to filter these results

Platform Type : **All**

Platform Type	Redirect URI	
Web	Edit	•••
Web	https://provider-vivacious-greyhound.ngrok-free.dev/auth/outlook/callback	

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Coda outlook | API permissions

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  - API permissions
  - Expose an API
  - App roles
  - Owners
  - Roles and administrators
  - Manifest
- Support + Troubleshooting

Refresh | Got feedback?

API / Permissions name	Type	Description	Admin consent requ...	Status	
Microsoft Graph (14)					•••
email	Delegated	View users' email address	No		•••
Mail.Read	Delegated	Read user mail	No	Granted for MohrStade	•••
Mail.Read.Shared	Delegated	Read user and shared mail	No		•••
Mail.ReadBasic	Delegated	Read user basic mail	No		•••
Mail.ReadBasic.Shared	Delegated	Read user and shared basic mail	No		•••
Mail.ReadWrite	Delegated	Read and write access to user mail	No		•••
Mail.ReadWrite.Shared	Delegated	Read and write user and shared mail	No		•••
Mail.Send	Delegated	Send mail as a user	No		•••
Mail.Send.Shared	Delegated	Send mail on behalf of others	No		•••
offline_access	Delegated	Maintain access to data you have given it access to	No	Granted for MohrStade	•••
openid	Delegated	Sign users in	No		•••
profile	Delegated	View users' basic profile	No		•••
User-Mail.ReadWrite.All	Delegated	Read and write secondary mail addresses for users	Yes	Not granted for MohrSt...	•••
User.Read	Delegated	Sign in and read user profile	No	Granted for MohrStade	•••

To view and manage consented permissions for individual apps, as well as your tenant's consent settings, try Enterprise applications.

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Coda outlook | Manifest

Search

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- Integration assistant
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- Manage
  - Branding & properties
  - Authentication (Preview)
  - Certificates & secrets
  - Token configuration
  - API permissions
  - Expose an API
  - App roles
  - Owners
  - Roles and administrators
  - Manifest
- Support + Troubleshooting

Add or remove favorites by pressing Ctrl+Shift+F

Refresh | Got feedback?

Starting from as early as 3/17/2025, and no later than 3/26/2025, you will no longer be able to view, save, upload, or download the Azure AD Graph app manifest on this page. Please follow the steps described in manifest migration guide to ensure a smooth migration.

An app manifest is a JSON representation of an app registration. The Microsoft Graph app manifest and AAD Graph app manifest below both represent the settings of this app registration. However they are in different formats. You only need to update one app manifest, and the system will automatically update the other one for them to match.

Microsoft Graph App Manifest (New) | AAD Graph App Manifest (Deprecating Soon)

See Understanding the Microsoft Graph application manifest to learn how to edit Microsoft Graph app manifest.

Save | Discard | Download | Upload

```
23    "samlMetadataUrl": null,
24    "defaultRedirectUri": null,
25    "certification": null,
26    "optionalClaims": null,
27    "requestSignatureVerification": null,
28    "addIns": [],
29    "api": {
30        "acceptMappedClaims": null,
31        "knownClientApplications": [],
32        "requestedAccessTokenVersion": 2,
33        "oauth2PermissionScopes": [],
34        "preAuthorizedApplications": []
35    },
36    "appRoles": [],
37    "info": {
38        "logoUrl": null,
```

Folder

OS (C:) › Users › Swarnim › zoom-outlook-backend07

Name	Date modified	Type	Size
node_modules	11/04/2026 12:23	File folder	
public	11/04/2026 12:47	File folder	
.env	11/04/2026 12:28	ENV File	1 KB
package	11/04/2026 12:23	JSON Source File	1 KB
package-lock	11/04/2026 12:23	JSON Source File	42 KB
server	11/04/2026 13:48	JavaScript Source ...	8 KB

OS (C:) › Users › Swarnim › zoom-outlook-backend07 › public

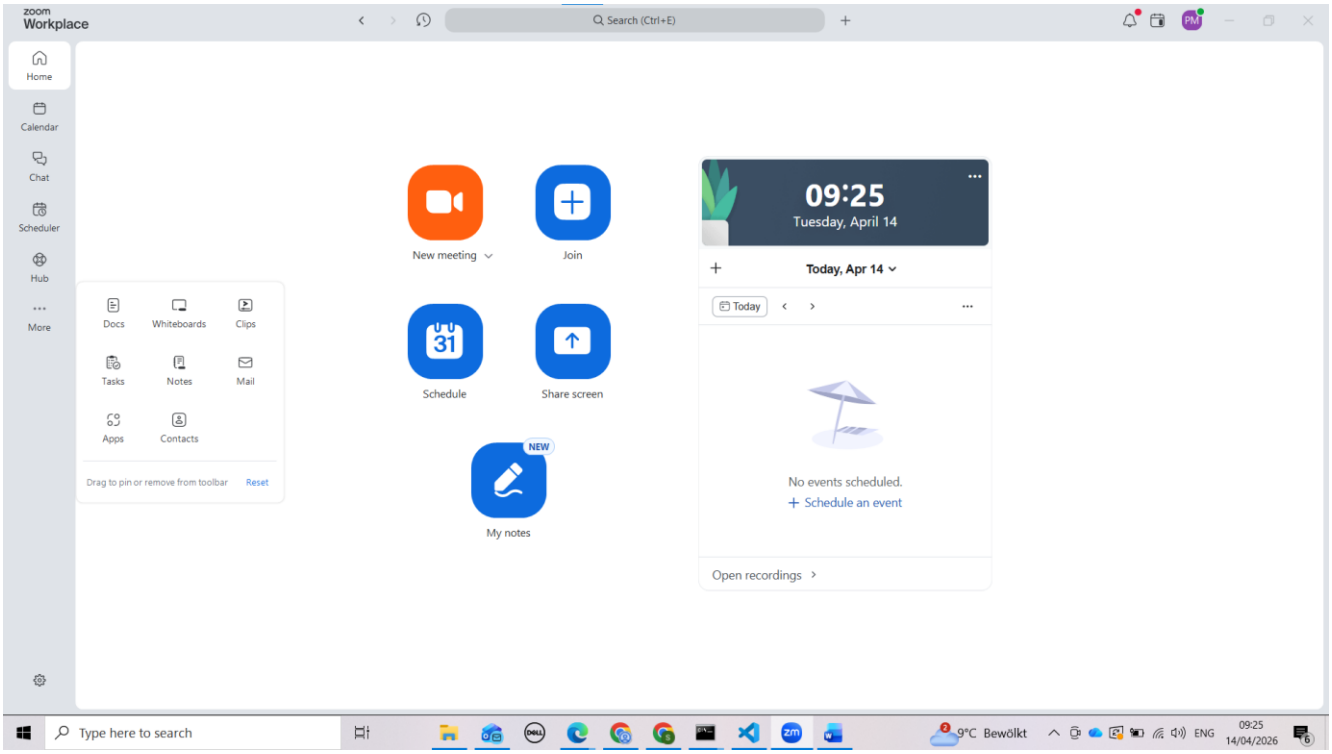
Name	Date modified	Type	Size
index	11/04/2026 13:48	Microsoft Edge HT...	8 KB

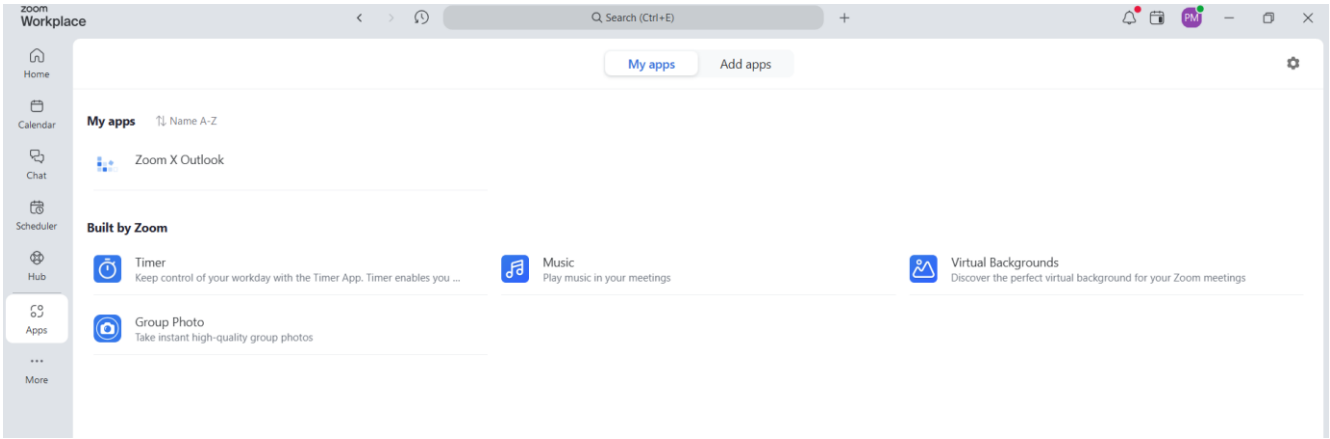
Command Prompt - ngrok http 3000

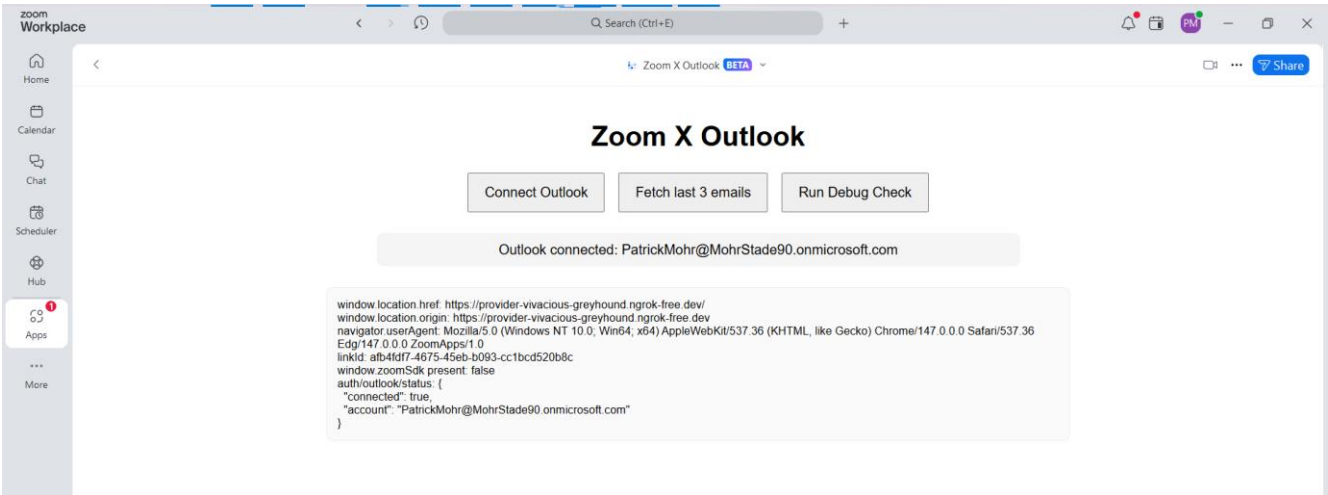
```
ngrok                                                                                                    (Ctrl+C to quit)

  One gateway for every AI model. Available in early access *now*: https://ngrok.com/r/ai

Session Status                online
Account                       swarnim@mohrstade.de (Plan: Free)
Update                        update available (version 3.37.6, Ctrl-U to update)
Version                       3.36.1-msix-stable
Region                        Europe (eu)
Latency                       12ms
Web Interface                 http://127.0.0.1:4040
Forwarding                    https://provider-vivacious-greyhound.ngrok-free.dev -> http://localhost:3000

Connections                   ttl     opn     rt1     rt5     p50     p90
                              62      0       0.00    0.00    6.28    10.31

HTTP Requests
-------------

09:20:56.411 CEST HEAD /                        200 OK
09:07:20.292 CEST HEAD /                        200 OK
10:06:41.938 CEST GET  /emails/latest           200 OK
10:06:37.814 CEST GET  /auth/outlook/callback   200 OK
10:06:35.675 CEST GET  /auth/outlook/status     304 Not Modified
10:06:34.315 CEST GET  /auth/outlook/start      302 Found
10:06:32.032 CEST GET  /zoom-sdk.js             304 Not Modified
10:06:32.224 CEST GET  /favicon.ico             404 Not Found
10:06:32.182 CEST GET  /auth/outlook/status     304 Not Modified
10:06:31.966 CEST GET  /                        304 Not Modified
```

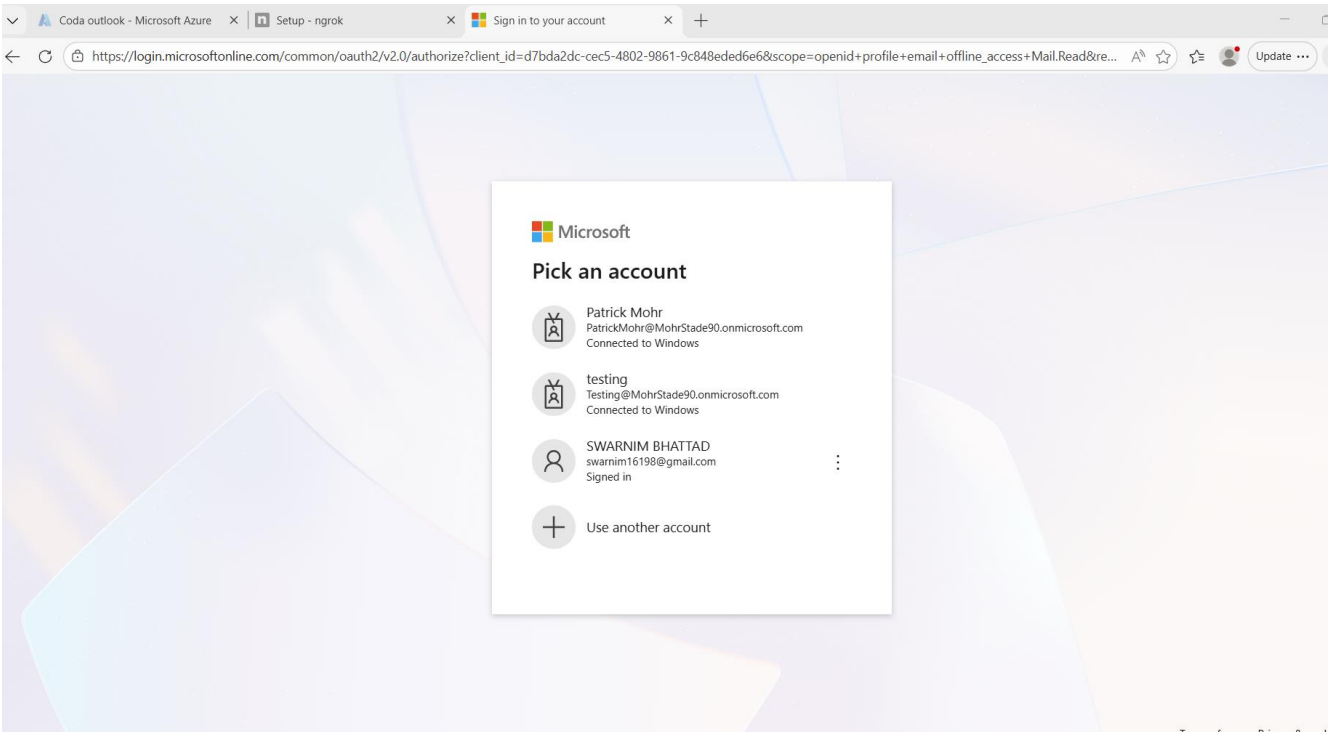
Command Prompt - node server.js

```
Server running on http://localhost:3000
^C
C:\Users\Swarnim\zoom-outlook-backend07>node server.js
 injected env (6) from .env // tip:  override existing { override: true }
Server running on http://localhost:3000
^C
C:\Users\Swarnim\zoom-outlook-backend07>node server.js
 injected env (6) from .env // tip:  auth for agents [www.vestauth.com]
Server running on http://localhost:3000
Zoom SDK proxy error: [TypeError: fetch failed] {
  [cause]: Error: read ECONNRESET
      at TLSWrap.onStreamRead (node:internal/stream_base_commons:216:20) {
    errno: -4077,
    code: 'ECONNRESET',
    syscall: 'read'
  }
}
Zoom SDK proxy error: [TypeError: fetch failed] {
  [cause]: Error: read ECONNRESET
      at TLSWrap.onStreamRead (node:internal/stream_base_commons:216:20) {
    errno: -4077,
    code: 'ECONNRESET',
    syscall: 'read'
  }
}
^C
C:\Users\Swarnim\zoom-outlook-backend07>node server.js
 injected env (6) from .env // tip:  secrets for agents [www.dotenvx.com]
Server running on http://localhost:3000
```


zoom
Workplace
Search (Ctrl+E)
Home
Calendar
Chat
Scheduler
Hub
More
Docs
Whiteboards
Clips
Tasks
Notes
Mail
Apps
Contacts
Drag to pin or remove from toolbar
Reset
New meeting
Join
Schedule
Share screen
NEW
My notes
09:25
Tuesday, April 14
Today, Apr 14
Today
No events scheduled.
Schedule an event
Open recordings
Type here to search
9°C Bewölkt
ENG
09:25
14/04/2026


zoom
Workplace
Search (Ctrl+E)
Home
Calendar
Chat
Scheduler
Hub
Apps
More
My apps
Add apps
My apps
Name A-Z
Zoom X Outlook
Built by Zoom
Timer
Keep control of your workday with the Timer App. Timer enables you ...
Music
Play music in your meetings
Virtual Backgrounds
Discover the perfect virtual background for your Zoom meetings
Group Photo
Take instant high-quality group photos





https://provider-vivacious-greyhound.ngrok-free.dev/auth/outlook/callback?code=1.AUEBft0rwnztSkmsOpybThk3_9yivdfFzgJlmGGchI7e1uYAAOFBAQ.BQABBAIAAAADAOz_BQD0_0...

Outlook connected successfully

Signed in as: PatrickMohr@MohrStade90.onmicrosoft.com

Connection ID: `afb4fdf7-4675-45eb-b093-cc1bcd520b8c`

You can now return to the Zoom app and click **Fetch last 3 emails**.

zoom Workplace

Search (Ctrl+E)

Home
Calendar
Chat
Scheduler
Hub
Apps
More

Zoom X Outlook BETA

Share

Zoom X Outlook

Connect Outlook | Fetch last 3 emails | Run Debug Check

Loaded 3 email(s)

Subject: (no subject)
From: Swarnim Bhattad <swarnim@mohrstade.de>
Received: 2026-04-14T07:00:53Z
Preview: Swarnim x Marcus Tuesday, April 14 · 9:00 – 9:30am Time zone: Europe/Berlin Google Meet joining info Video call link: https://meet.google.com/twq-sdeb-qkx Or dial: (DE) +49 40 8081615975 PIN: 722 442 644# More phone numbers: https://tel.meet/t

Subject: Ihre größten KI-Fragen, beantwortet
From: Use AI <hello@use.ai>
Received: 2026-04-13T16:32:53Z
Preview: Hallo, Sie sind nicht die Ersten, die sich fragen, ob KI wirklich Ihre tägliche Produktivität steigern kann. Jetzt ausprobieren Hier sind einige Fragen, die andere vor dem Start mit Use AIstellten, und was sie entdeckten: F: Wie viel Zeit spar

Subject: Du wirst es öfter nutzen, als du denkst!
From: Daniel from Use AI <hello@use.ai>
Received: 2026-04-12T16:32:53Z
Preview: Hallo, Das ist zwar etwas unangenehm, aber wir müssen fragen: Was hält Sie davon ab, Use AI zu nutzen? Sind Sie sich nicht sicher, ob Sie es tatsächlich nutzen werden? Denn das haben wir von über 50.000 Mitgliedern gelernt: Der größte Bedauern